

# 1 Introduction

The single well 1D harmonic oscillator is one of the most fundamental and commonly solved problems in undergraduate quantum mechanics courses. Traditionally, in most introductory quantum mechanics textbooks, the governing differential equation is solved analytically using a power series method, which ultimately leads to the Hermite polynomials, or algebraically using ladder operators. In fact, one or both of these methods were the exclusive approach taken in the more than thirty texts we reviewed for this study where the solution for the 1D quantum oscillator was given and discussed [1–39]. However, we demonstrate here a less laborious route to the complete solution for this problem. By employing one straightforward variable transformation, the governing equation can be written in the form of a parabolic cylinder equation, and the resulting state functions are thus given in terms of the parabolic cylinder functions. In the list of texts mentioned above, only Merzbacher [22] utilized a parabolic cylinder equation solution for any oscillator problem, and this is the double harmonic oscillator, not the single well 1D oscillator that we consider here.

Additionally, a similar approach can be used to solve the Schrödinger equation for the 1D harmonic oscillator in a uniform electric field. In this case, the resulting parabolic cylinder equation reveals the possibility of two solutions. One is the well-known result where the 1D oscillator eigenvalues are reduced by a frequency-dependent term that can have any positive value [17]. The other is where the field term is restricted to be an integer, and the eigenvalues are more like those for the field-free case.

In the next section, steps are given for arriving at the solution for the 1D harmonic oscillator in terms of parabolic cylinder functions. In the subsequent section, the case where the oscillator is within a uniform electric field is considered. Again, steps leading to the solution in terms of parabolic cylinder functions are given. Finally, we demonstrate how the results for the oscillator in a uniform electric field can be used to approximate the bound states of a Lennard-Jones potential.

## 2 Solutions to the Quantum Harmonic Oscillator

We consider the classical potential energy,  $V$ , for the one dimensional (1D) harmonic oscillator as,  $V = \frac{1}{2}\mu\omega^2x^2$ , where  $\mu$  is reduced mass,  $k$  is the force constant and  $x$  is position. It is useful to write the force constant in terms of the oscillator's natural frequency  $\omega$  as  $\omega = \sqrt{k/\mu}$ . With this substitution, the full Hamiltonian is written as

$$\hat{H} = -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2 . \quad (1)$$

Therefore, the time-independent Schrödinger equation for the 1D harmonic oscillator is:

$$-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2}\psi + \frac{1}{2}\mu\omega^2x^2\psi = E\psi . \quad (2)$$

After a bit of manipulation this becomes

$$\frac{\hbar}{2\mu\omega} \frac{d^2}{dx^2}\psi + \left( \frac{E}{\hbar\omega} - \frac{\mu\omega}{2\hbar}x^2 \right) \psi = 0 . \quad (3)$$